Robust Learning Under Label Noise for Real-World Classification

Evaluation & Reproducibility Report | Experiment ID: 697d888f-9a56-4ea3-bc65-6e6e55b0dceb | Generated: 2026-10-07 00:45:23

1. Dataset & Experimental Setup

Target Variable:	cultivar	**Classes:**	3 (Cultivar_A, Cultivar_B, Cultivar_C)
Total Rows:	178	**Preprocessed Features:**	13
Train / Test Split:	124 (70%) / 27 (15%)	**Noise Levels Evaluated:**	0%, 10%, 20%, 30%
Random Seed:	[42]	**Noise Strategy:**	Symmetric

2. Model Evaluation Across Noise Levels

Evaluated strictly on the clean holdout test set to measure genuine generalization ability under synthetic training noise.

Model	Type	Noise	Accuracy	Precision	Recall	F1-Score	Time (s)
Baseline Logistic Regression	Baseline	0%	1.0000	1.0000	1.0000	**1.0000**	0.06
Baseline Random Forest	Baseline	0%	1.0000	1.0000	1.0000	**1.0000**	1.15
Robust Sample-Weighted (Random Forest)	Robust	0%	1.0000	1.0000	1.0000	**1.0000**	0.88
Robust Trimmed (Random Forest)	Robust	0%	1.0000	1.0000	1.0000	**1.0000**	0.49
Baseline Logistic Regression	Baseline	10%	1.0000	1.0000	1.0000	**1.0000**	0.02
Baseline Random Forest	Baseline	10%	1.0000	1.0000	1.0000	**1.0000**	0.38
Robust Sample-Weighted (Random Forest)	Robust	10%	1.0000	1.0000	1.0000	**1.0000**	0.33
Robust Trimmed (Random Forest)	Robust	10%	1.0000	1.0000	1.0000	**1.0000**	0.30
Baseline Logistic Regression	Baseline	20%	0.8148	0.8350	0.8148	**0.8160**	0.02
Baseline Random Forest	Baseline	20%	1.0000	1.0000	1.0000	**1.0000**	0.42
Robust Sample-Weighted (Random Forest)	Robust	20%	1.0000	1.0000	1.0000	**1.0000**	0.61
Robust Trimmed (Random Forest)	Robust	20%	1.0000	1.0000	1.0000	**1.0000**	0.65
Baseline Logistic Regression	Baseline	30%	0.7778	0.7904	0.7778	**0.7794**	0.02
Baseline Random Forest	Baseline	30%	0.8519	0.8584	0.8519	**0.8488**	0.48
Robust Sample-Weighted (Random Forest)	Robust	30%	0.8148	0.8181	0.8148	**0.8108**	0.47
Robust Trimmed (Random Forest)	Robust	30%	1.0000	1.0000	1.0000	**1.0000**	0.51

3. Confident Learning Label Issue Detection

Cross-validation out-of-fold probability estimation identified **24** suspicious training samples (19.35% of training split) exceeding the issue threshold of 0.50. These samples exhibited substantial margin disagreement between the assigned label and the learned distribution.

Row ID	Observed Label	Predicted Class	Confidence	Observed Prob	Issue Score	Status
33	Cultivar_B	Cultivar_A	0.99	0.01	**0.99**	Potential Issue
0	Cultivar_B	Cultivar_A	0.94	0.05	**0.92**	Potential Issue
31	Cultivar_C	Cultivar_A	0.88	0.02	**0.92**	Potential Issue
92	Cultivar_C	Cultivar_B	0.89	0.03	**0.92**	Potential Issue
168	Cultivar_B	Cultivar_C	0.93	0.05	**0.91**	Potential Issue
144	Cultivar_A	Cultivar_C	0.83	0.04	**0.87**	Potential Issue
74	Cultivar_A	Cultivar_B	0.90	0.08	**0.87**	Potential Issue
114	Cultivar_A	Cultivar_B	0.88	0.08	**0.86**	Potential Issue

4. Academic Findings & Discussion

• **Baseline Performance:** Under pristine labels (0% noise), Baseline Logistic Regression achieved the highest F1-score of 1.0000 (Accuracy: 1.0000).

• **Noise Robustness:** At 30% label noise, the robust model (Robust Trimmed (Random Forest)) demonstrated superior noise resilience, achieving an F1-score of 1.0000 compared to Baseline Random Forest's 0.8488 (+0.1512 improvement).

• **Noise Degradation:** Baseline Logistic Regression experienced an F1 degradation of 0.2206 as noise increased from 0% to 30%.

5. Methodological Notes & Reproducibility

Reproducibility: Results are strictly deterministic given the configured random seed and split ratios. Label noise was applied solely to the training set; test labels remained unaltered to ensure valid evaluation.
Academic Note: Confident learning scores indicate statistical discrepancy between observed labels and feature density; they represent potential label noise, not mathematical proof of label error in uncontrolled environments.